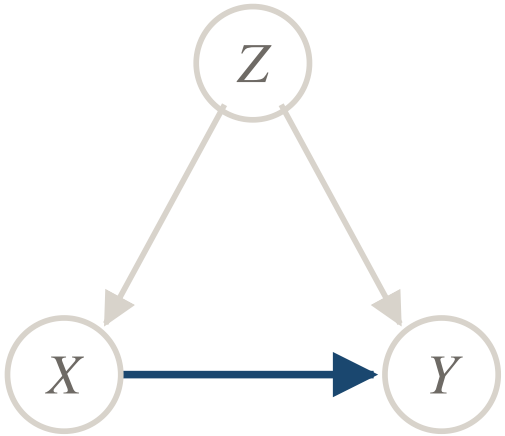
Controls and Designs as Causal Claims

TEACHING NOTE

ACCTG 597A · Week 8

On Gow, Larcker, and Reiss (2016), "Causal Inference in Accounting Research," *Journal of Accounting Research* 54(2): 477–523. Running examples, as GLR discuss them: Larcker, Richardson, and Tuna (2007) and Armstrong, Gow, and Larcker (2013).



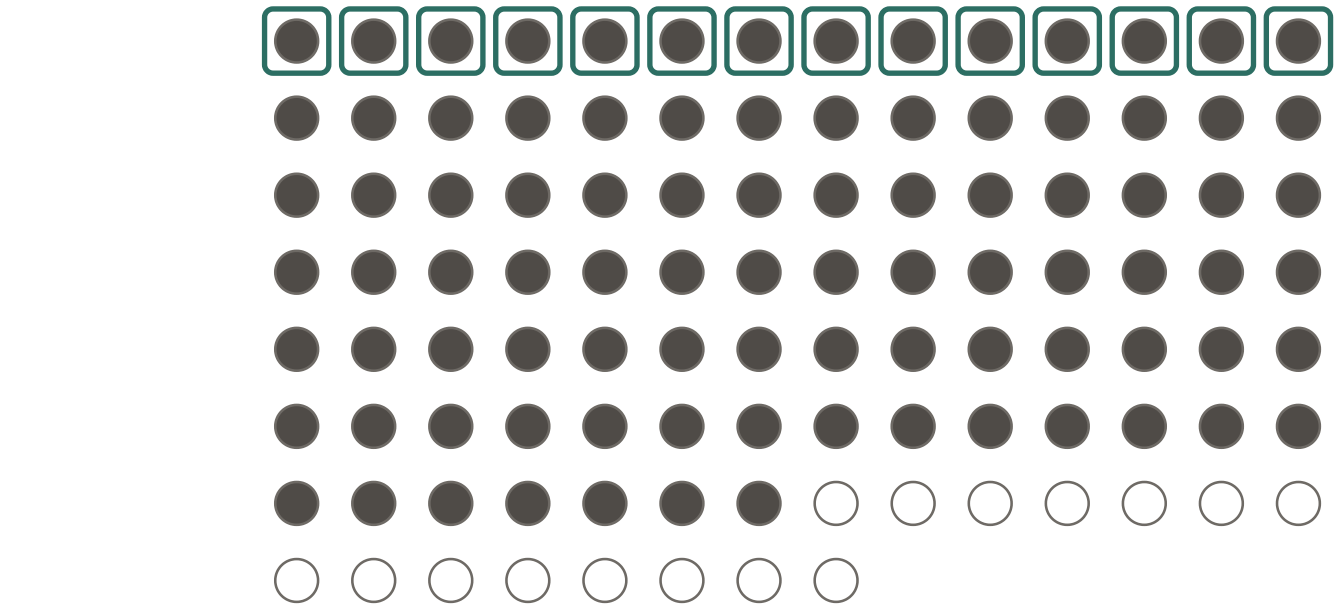
GLR coded every 2014 paper in JAR, TAR, and JAE

Of 139 papers, 125 report original research and 106 of those use observational data. For each of the 106, GLR asked whether the paper sought to draw a causal inference.

The room's estimate, before the circles fill: how many of the 106 sought a causal inference.

- **Of the 106, 91 sought a causal inference; 14 of the 91 used a quasi-experimental design, and the rest rely on regression or matching controls.**

> *"While it is widely understood that OLS regressions that use observational data produce unbiased estimates of causal effects only under very strong assumptions, the credibility of these assumptions is rarely explicitly addressed."*
>
> Gow, Larcker, and Reiss (2016, p. 481)



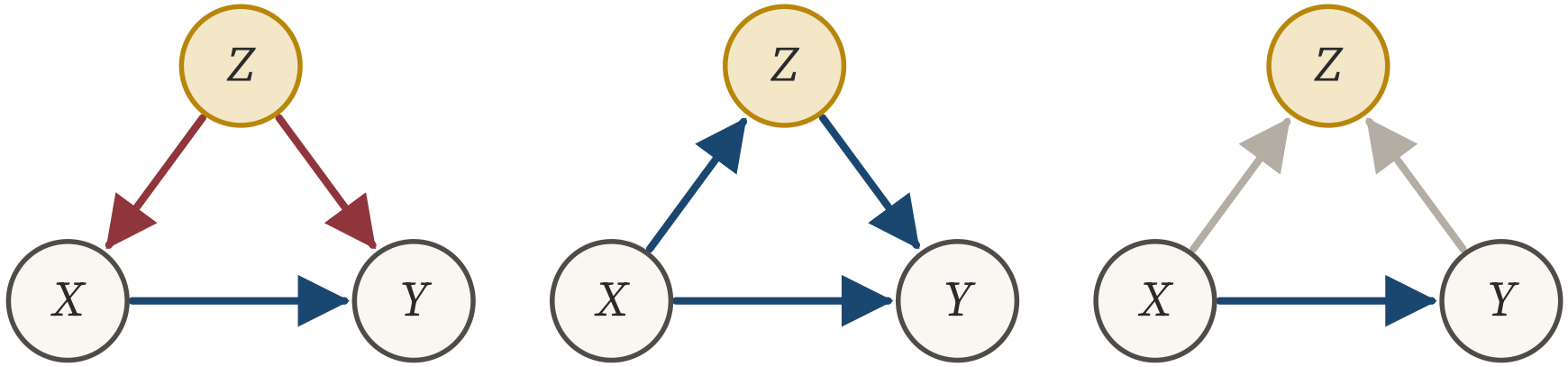
sought a causal inference: 91 of 106 · description, prediction, or measurement: 15 · quasi-experimental design: 14 of the 91

One circle per observational paper in GLR's 2014 census (pp. 480–481, 485); the counts are in GLR's text, with no table. 91 of 106 is 86 percent, which GLR call about 90 percent; the other 15 are description (7), prediction (5), and measurement (3). The teal ring marks the 14 quasi-experimental papers, 5 natural experiments and 10 IV papers with one paper in both lists.

A causal diagram states which arrows are present

- **In GLR's Figure 1 the same control Z is a confounder, a mediator, or a collider depending only on the direction of two arrows; the data cannot tell the three apart, and a theory of the institution can.**

Running example (GLR pp. 484–485): Larcker, Richardson, and Tuna (2007) regress firm outcomes and performance on governance factors, with and without a list of controls whose roles are not stated.



(A) Z is a confounder **(B) Z is a mediator** **(C) Z is a collider**

Redrawn from GLR Figure 1 (p. 483): Treatment variable (X), Outcome variable (Y), and "Control" (Z). Navy: the causal path. Maroon: an open back-door path, a path between X and Y that starts with an arrow into X (GLR fn. 9, p. 484). Grey: a path closed at Z. Throughout the deck a dashed arrow with a hollow head is one assumed absent, and a teal ring marks a variable held fixed.

> *"Statistics can help us determine whether X and Y are correlated, but correlations do not establish causality. Only with assumptions about causal relations between X, Y, and other variables (i.e., a theory) can we infer causality."*
>
> Gow, Larcker, and Reiss (2016, p. 482)

Controlling for a confounder removes the bias

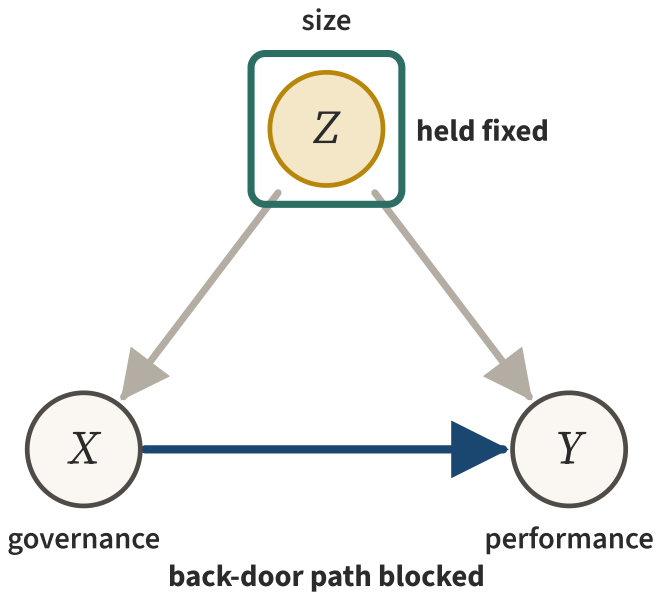
- **When a control drives both governance and performance, the uncontrolled slope mixes the effect with the back-door path, and conditioning on Z returns the effect.**

GLR's linear case (p. 518), with the omitted-variable slope written out here

$$\text{plim } b_{Y\cdot X} = \beta + \gamma\,\delta, \qquad \delta = \text{slope of } Z \text{ on } X$$

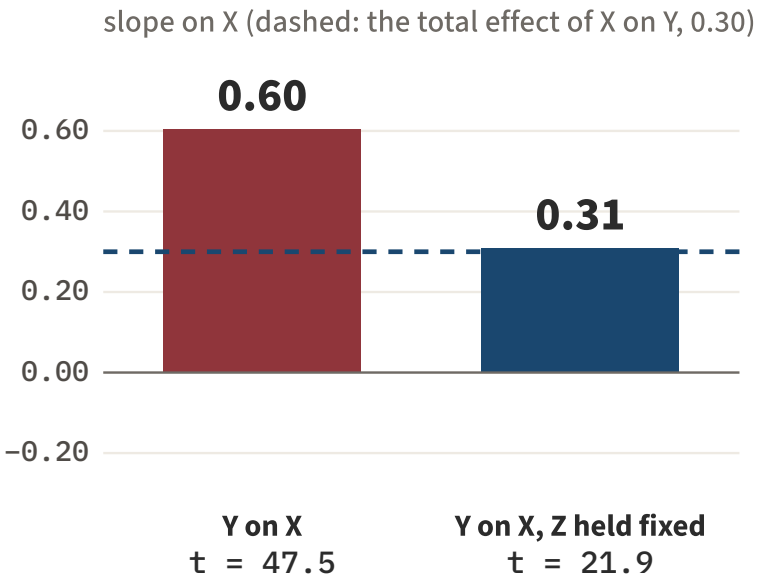
Here $\beta = 0.30, \gamma = 0.50, \delta = 0.6$: the slope without Z is **0.60** in population and 0.60 in the draw; with Z, 0.31.

Instance for this note (GLR name none for this role): firm size, which shapes board structure and performance alike. The identity $b_{Y\cdot X} = b_{Y\cdot XZ} + \hat{\gamma}\,\hat{\delta}$ is exact in every sample. Appendix: the four closed forms.



Control for size (Z)

Slope of Y on X holding Z fixed: 0.31 (t = 21.9), the causal effect.



Simulated (seed 2016, N = 5,000): Z ~ N(0, 1), X = 0.6 Z + u, Y = 0.30 X + 0.50 Z + e. The causal effect of X on Y is 0.30 by construction. Bars: one draw; t-statistics below.

Controlling for a mediator changes the question

- **If governance works partly through the control, holding it fixed reports the direct effect in place of the total effect.**

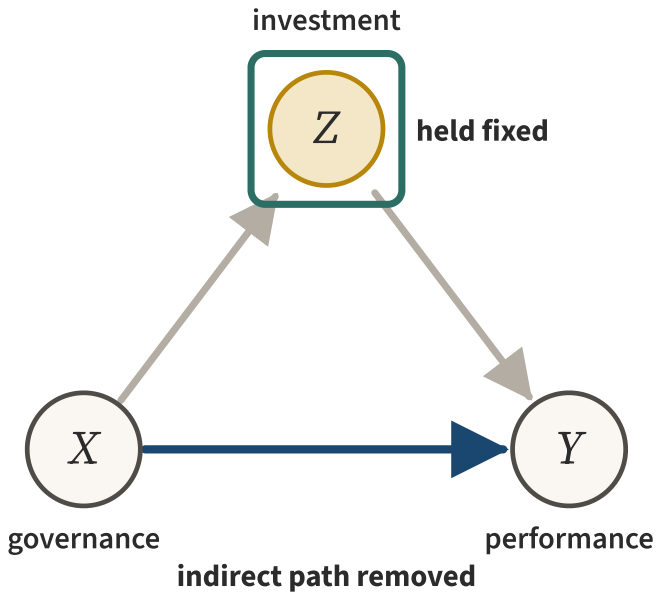
Week 6: the total effect is the direct effect plus the indirect effect

$$c = c' + a\,b, \qquad 0.30 = 0.10 + 0.5 \times 0.4 \text{ in population}$$

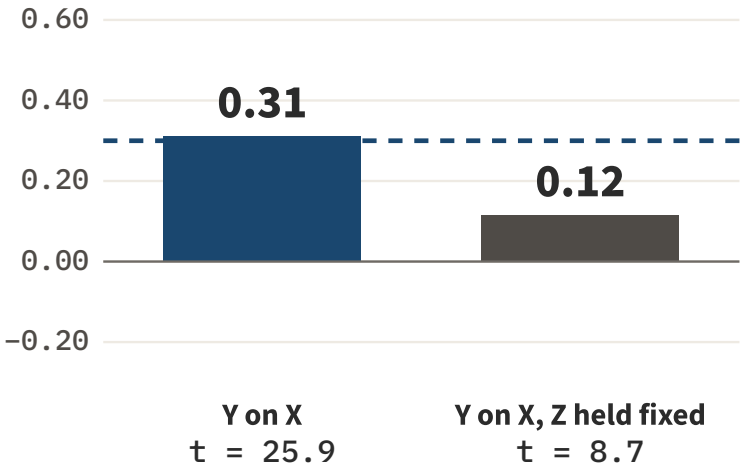
Larcker, Richardson, and Tuna call their with-controls estimates "conservative" because governance is assumed to have no impact on the controls (quoted by GLR, pp. 484–485). GLR's reply:

> *"the claim that the estimates are "conservative" is only correct if the indirect effect via mediators is of the same sign as the direct (i.e., unmediated) effect."*
>
> Gow, Larcker, and Reiss (2016, p. 485)

Instance for this note: an investment policy that governance sets and that moves performance.



slope on X (dashed: the total effect of X on Y, 0.30)



Control for investment (Z)

Slope of Y on X holding Z fixed: `0.12` (t = 8.7), the direct effect only.

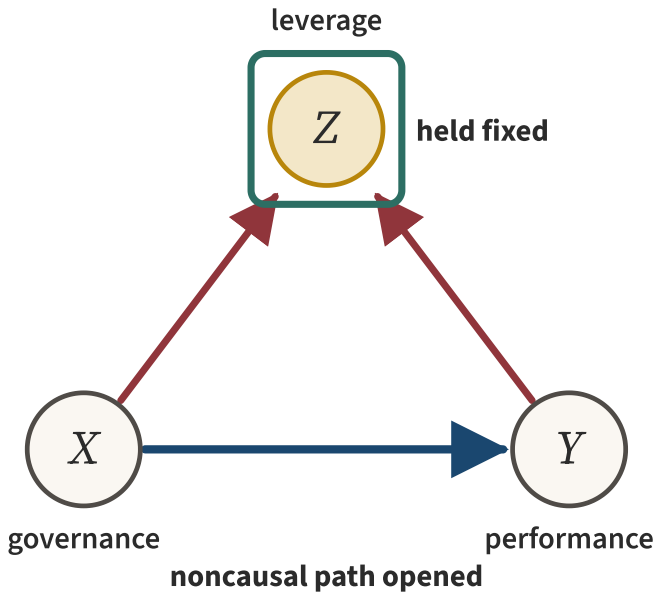
Simulated (seed 2016, N = 5,000): X ~ N(0, 1), Z = 0.5 X + u, Y = 0.10 X + 0.40 Z + e. The total effect of X on Y is 0.30 by construction; the direct effect is 0.10. Bars: one draw; t-statistics below.

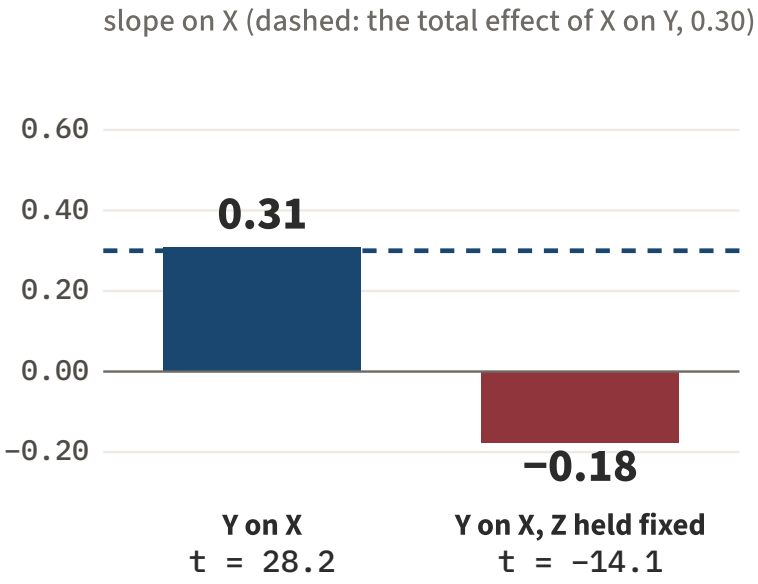
Conditioning on a collider induces a spurious association

- **Leverage responds to both governance and performance; holding it fixed opens a noncausal path between them and biases the slope.**

> *"governance plausibly affects leverage choices, while performance is also likely to affect leverage. If so, "controlling for" leverage might induce associations between governance and performance even absent in a true relation between these variables."*
>
> Gow, Larcker, and Reiss (2016, p. 485)

GLR's instance (p. 485; their fn. 12 notes that LRT do not in fact control for leverage when performance is the outcome). Closed form in the appendix, the four closed forms: the population slope given Z is −0.201.





Control for leverage (Z)

Slope of Y on X holding Z fixed: −0.18 (t = −14.1), biased: conditioning opened the path.

Simulated (seed 2016, N = 5,000): X ~ N(0, 1), Y = 0.30 X + e, Z = 0.7 X + 0.7 Y + v. The causal effect of X on Y is 0.30 by construction. Bars: one draw; t-statistics below.

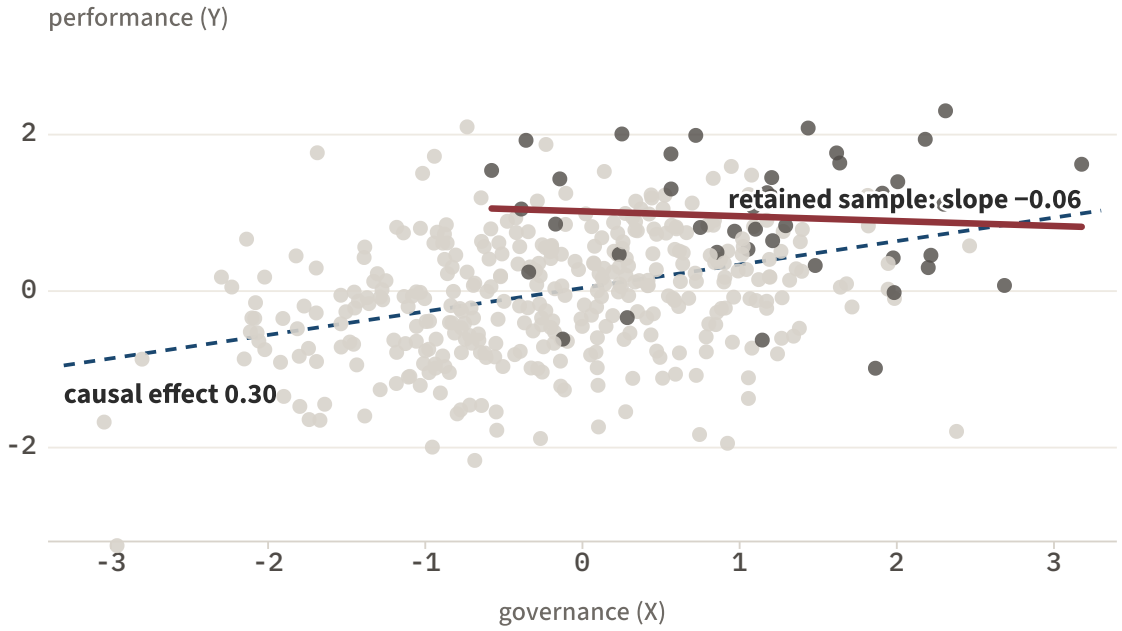
Selecting the sample is conditioning on a collider

- **Keeping only the firms with the highest leverage, the same collider as before, bends the retained-sample slope from 0.31 toward zero and below without any control being added.**

> *"Many intuitive examples of collider bias involve selection or stratification."*
>
> Gow, Larcker, and Reiss (2016, p. 484)

Samples restricted to firms that are covered, rated, listed, or still alive are selected on variables the treatment and the outcome may both affect (a reading for this note; GLR's example is college admission, p. 484).

Slopes at each retained share come from the full retained sample (seed 2016).



Retained by leverage — top 10%

Keep the top 10% by leverage (N = 500 of 5,000): slope of Y on X −0.06 (t = −1.6).

The collider draw of the previous slide (seed 2016): Y = 0.30 X + e and Z = 0.7 X + 0.7 Y + v. Dots: 400 of the 5,000 observations; slopes from the full retained sample. Grey dots are dropped.

Each control needs a stated role

- **A control enters a regression with a claim: it is a confounder, and it is neither a mediator nor a collider; GLR observe that accounting papers seldom make the claim explicit.**

Role of Z	Arrows	Rule (GLR, pp. 484, 518)	Slope of Y on X: without Z, with Z	An instance (this note's, except leverage)
Confounder	Z → X, Z → Y	condition on Z	0.60, 0.31	firm size, shaping board structure and performance alike
Mediator	X → Z → Y	do not, for the total effect	0.31, 0.12 (direct effect)	an investment policy governance sets and that moves performance
Collider	X → Z ← Y	do not	0.31, −0.18	leverage (GLR, p. 485); a sample kept by coverage, rating, or survival

> *"While many of these variables should be considered confounders, less attention is given to explaining why it is reasonable to assume that they are not mediators or colliders. Such a discussion is important because the inclusion of "controls" that are mediators or colliders will generally lead to bias."*
>
> Gow, Larcker, and Reiss (2016, p. 484)

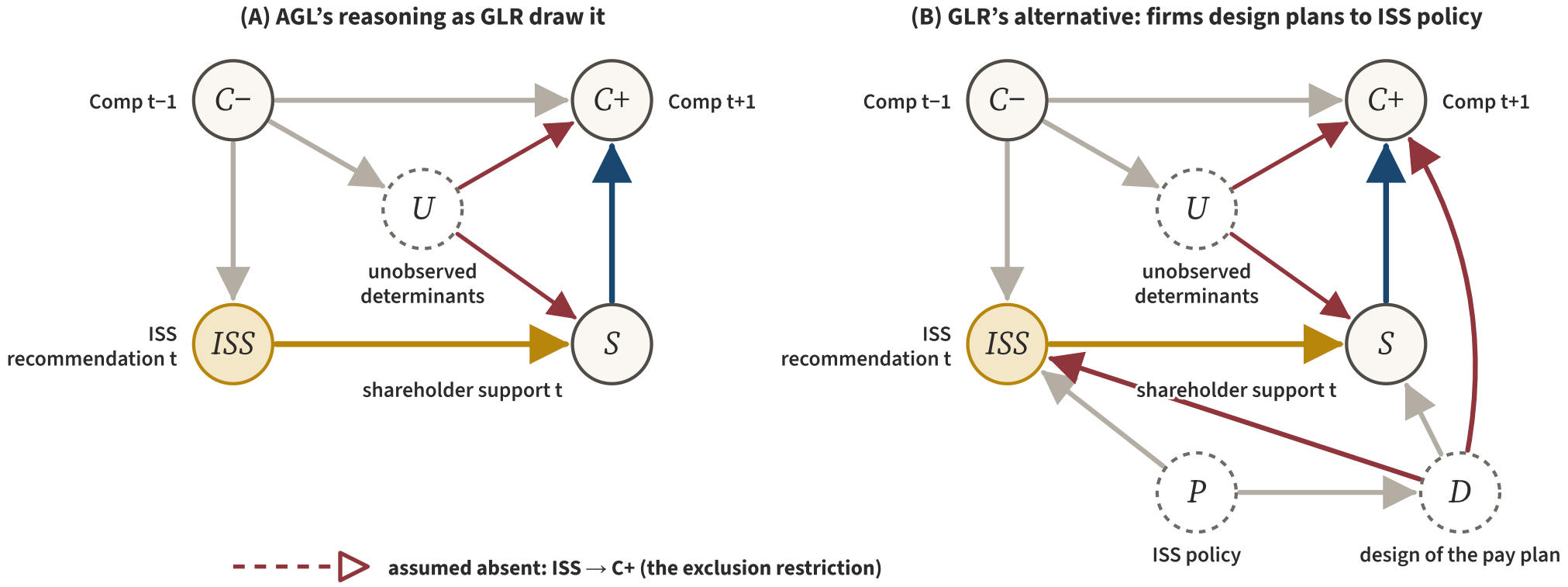
How to tell, for this note: a control measured before the treatment cannot be its mediator or a collider of the treatment and the outcome; whether it is a confounder is a claim about the institution. GLR's rule (Definition 2, p. 517): a conditioning set contains no descendant of X and blocks every path between X and Y that starts with an arrow into X. Appendix: the criterion, formally.

A quasi-experimental design is a claim that certain arrows are absent



An instrument is valid only if two kinds of arrow are absent

- Armstrong, Gow, and Larcker (2013) ask whether shareholder support (S) changes next year's pay (C+), instrumenting support with the ISS recommendation; the instrument needs no arrow from the recommendation to pay and no cause shared with pay, and GLR's alternative diagram supplies a shared cause.



Redrawn from GLR Figure 2 (p. 491); dashed nodes are unobserved. Gold: the instrument and the first stage; navy: the effect of interest. Maroon: open noncausal paths, in (A) the confounding of S and C+ by U that motivates the instrument, in (B) the path ISS ← D → C+ that plan design opens between instrument and outcome. The dashed, hollow-headed arrow under (A) is the one GLR do not draw: the exclusion restriction.

> *"This assumption is represented in figure 2(A) by the absence of an arrow from ISS recommendation$_t$ to Comp$_{t+1}$."*
>
> Gow, Larcker, and Reiss (2016, p. 519)

A weak instrument magnifies a small violation

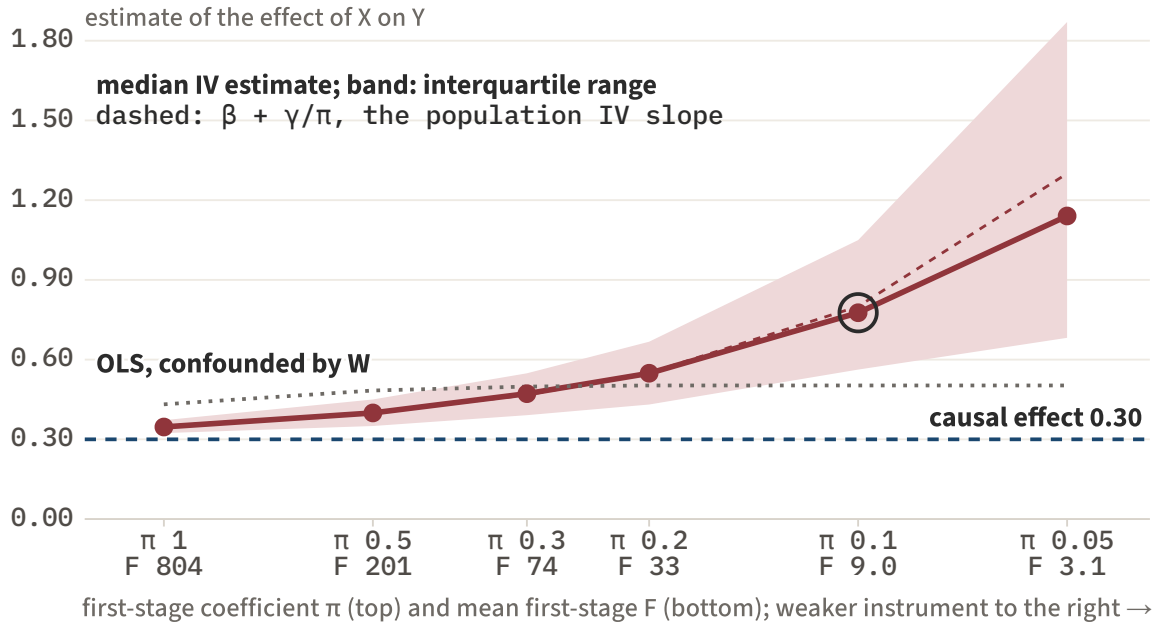
- **With a direct effect of the instrument one sixth of β, IV is 0.35 at F ≈ 800 but 0.78 at F ≈ 9, further from 0.30 than OLS at 0.50; a cause shared with the outcome enters the same way.**

Any covariance between the instrument and the structural error, over the first stage

$$\text{plim } b_{IV} = \beta + \frac{\text{Cov}(Z,u)}{\text{Cov}(Z,X)} = \beta + \frac{\gamma}{\pi} \text{ when } u \text{ contains } \gamma Z$$

> *"no statistical test allows the researcher to verify that their instruments satisfy the exclusion restriction."*
>
> Gow, Larcker, and Reiss (2016, p. 489)

The simulation is this note's; γ is the arrow absent from GLR's Figure 2(A). GLR's own simulation (appendix: GLR's simulation) uses strong but spurious instruments, and GLR find that the 2014 IV papers "have paid little heed" to Larcker and Rusticus (2010) (p. 491). Appendix: the four closed forms.



First stage ——●—— π = 0.1

π = 0.1, mean first-stage F 9.0: median IV 0.78 [0.56, 1.05]; OLS 0.50; causal effect 0.30; population IV 0.80.

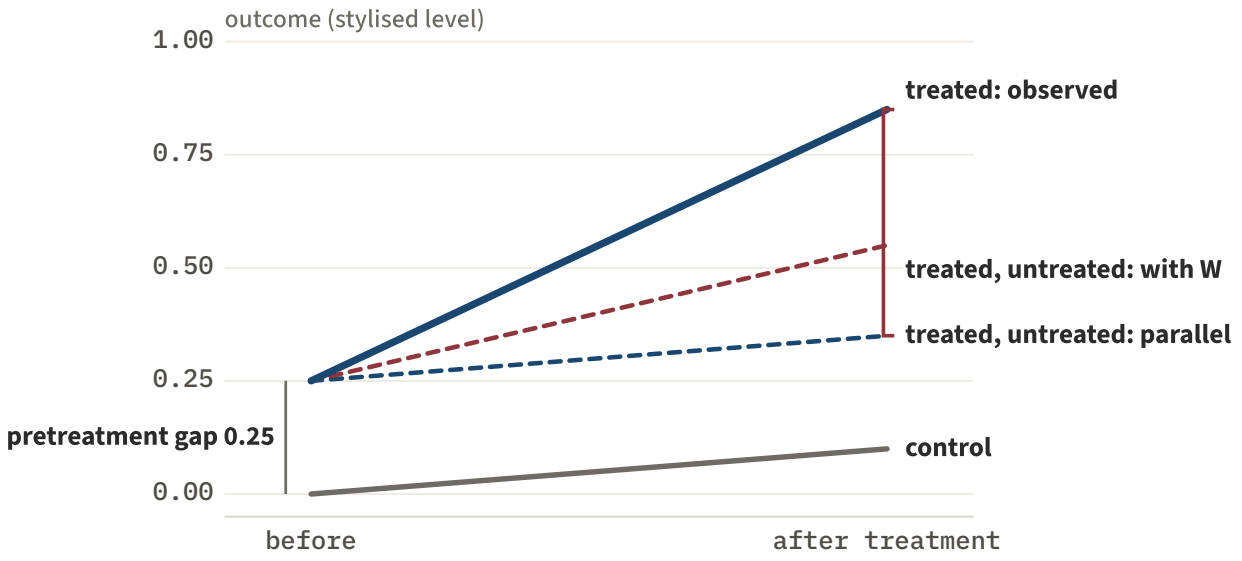
Monte Carlo (seed 2016): 1,000 draws of N = 1,000 per point. X = π Z + 0.5 W + v and Y = 0.30 X + 0.05 Z + 0.5 W + e, with W unobserved. The instrument's direct effect γ = 0.05 is one sixth of β.

Parallel trends is the absence of a confounder of the change

- **Differencing removes a pretreatment gap, not a W that both selected the treated group and moves their change; GLR read such a gap as evidence against as-if random assignment, and, where outcomes mean-revert, as a sign of such a W.**

> *"If treatment assignment is driven by unobserved confounding variables, then DD and fixed-effect estimates of causal effects will be biased and inconsistent."*
>
> Gow, Larcker, and Reiss (2016, p. 493)

Kelly and Ljungqvist (2012), as GLR read them (p. 493, fn. 27): shocks to brokerage coverage, a DiD on spreads, and a 0.039 pretreatment difference in spreads between treated and matched control firms, nearly twice the estimated effect of 0.020. Appendix: what GLR leave out.



W-driven change of the treated group 0.20

DiD estimate = (treated change 0.60) − (control change 0.10) = 0.50; the causal effect is 0.30; the pretreatment gap 0.25 is removed by differencing.

Stylised two-period picture (this note's), drawn from the closed form $\tau + \theta\ \mathrm{Cov}(D, W)/\mathrm{Var}(D)$ with $\tau = 0.30$; the slider sets the second term, the change in the outcome that W adds to the treated group. Levels are illustrative.

The other designs make the same kind of claim

- **Each design replaces a list of controls with one stated absence; the absence is an assumption about the institution, not a property of the estimator.**

Design	The arrow assumed absent	GLR's admonition
Natural experiment	none into treatment from anything that also reaches the outcome	assignment is not treatment: a randomly assigned drug can appear to raise blood pressure (p. 487); "credible natural experiments are very rare" (p. 488)
Regression discontinuity	none at the cutoff: no manipulation of the running variable	the estimate is local (a zero at the SOX $75 million threshold is consistent with large effects elsewhere); test for manipulation, choose the bandwidth, plot the data (p. 492)
Propensity-score matching	none from unobservables into treatment	matching "does not necessarily eliminate the endogeneity problem resulting from unobservable variables" (Minutti-Meza 2014, quoted p. 494)
Fixed effects	none from time-varying unobservables into treatment	effects of mixed sign, always-treated units dropped; when time-invariant heterogeneity is correlated with potential outcomes, fixed effects "can have greater bias than estimators that omit fixed effects" (p. 493)

> *"it is the detailed understanding of the research setting, not the method per se, that makes these estimates credible."*
>
> Gow, Larcker, and Reiss (2016, fn. 6, p. 481)

GLR's paths forward all establish which arrows exist

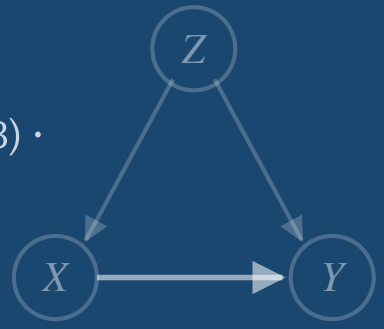
- **Statistical methods alone cannot establish the arrows; GLR ask for causal mechanisms, descriptive research on institutions, and structural models, and for the diagram as the minimal written statement of what is assumed.**

1. **Mechanisms.** Snow's cholera natural experiment was accepted only after Koch isolated the organism (pp. 495–496).
2. **Descriptive research.** Its test: whether the results would change a research agenda or a causal interpretation of prior work (p. 501).
3. **Structural models.** GLR's illustration: on simulated data in which equity pay raises the benefit from misstating, a logit gives equity pay a negative coefficient, because a strategic auditor works harder when incentives are stronger; the structural model recovers the positive sign (Tables 2 and 3). Appendix: the strategic auditor.

> *"One tool that may help in this regard are causal diagrams because these require the research to be very transparent about the causal mechanism that is being assumed for the research question."*
>
> Gow, Larcker, and Reiss (2016, p. 516)

Every control and every design is a causal claim

1. A control variable enters a regression with a claim about arrows: a confounder removes a bias, a mediator changes the estimand, and a collider manufactures an association. The data do not distinguish the three roles, so the claim has to be stated and defended from the institution.
2. A quasi-experimental design restates the claim as an absence: an exclusion restriction, parallel trends, or no manipulation at a cutoff. No statistical test verifies an absent arrow, and a weak first stage magnifies a small violation.
3. The census that opened this note counted causal inferences, not stated assumptions. The question GLR leave with each reader is which row of the three-row table each control in a current project sits in, and what in the institution establishes it; the diagram is the written form of that answer.



Gow, Larcker, and Reiss (2016) · Larcker, Richardson, and Tuna (2007) · Armstrong, Gow, and Larcker (2013) · github.com/sbb151/acctg597a-research-design

Appendix



The back-door criterion, formally

- **GLR's appendix states the rule the three stages illustrate: a conditioning set must block every back-door path into X and contain no descendant of X.**

> *"DEFINITION 2. (Back-door criterion). A set of variables Z satisfies the back-door criterion relative to an ordered pair of variables (X, Y) in a directed acyclic graph (DAG) G if [...] no node in Z is a descendant of X, and [...] Z blocks every path between X and Y that contains an arrow into X."*
>
> Gow, Larcker, and Reiss (2016, p. 517); the bracketed ellipses replace the bullet markers

A path is blocked (Definition 1, p. 517) when it contains a chain $i \to m \to j$ or a fork $i \leftarrow m \to j$ with m conditioned on, or a collider $i \to m \leftarrow j$ with neither m nor a descendant of m conditioned on. Theorem 1 (p. 517) then gives the adjustment formula $P(y \mid \hat{x}) = \sum_z P(y \mid x, z)\, P(z)$.

Applied to Figure 1 (p. 518): in (A) the set {Z} blocks X ← Z → Y; in (B) and (C) Z is a descendant of X, and the empty set satisfies the criterion, so one "should not condition on Z."

Back

Four closed forms the stages draw

- **Each stage draws a population slope with a closed form; the simulation checks the finite-sample identities and the symbolic file checks the algebra.**

Confounder, $Z \sim (0, 1)$, $X = aZ + u$, $Y = \beta X + \gamma Z + e$, with $a = 0.6$, $\sigma_u^2 = 0.64$, $\beta = 0.30$, $\gamma = 0.50$:

$$\operatorname{plim}\, b_{Y \cdot X} = \beta + \gamma \, \frac{a}{a^2 + \sigma_u^2} = 0.30 + 0.50 \times \frac{0.6}{1.0} = 0.60$$

Mediator, $Z = aX + u$, $Y = c'X + bZ + e$, with $a = 0.5$, $b = 0.4$, $c' = 0.10$:

$$b_{Y \cdot X} = c' + ab = 0.10 + 0.5 \times 0.4 = 0.30, \qquad b_{Y \cdot XZ} = c' = 0.10$$

Collider, $Y = \beta X + e$, $Z = \delta_X X + \delta_Y Y + v$, with $\delta_X = \delta_Y = 0.7$, $\sigma_e^2 = 0.64$, $\sigma_v^2 = 0.5$:

$$b_{Y \cdot XZ} = \beta - \frac{\delta_Y (\delta_X + \delta_Y \beta)\, \sigma_e^2}{\delta_Y^2 \sigma_e^2 + \sigma_v^2} = 0.30 - \frac{0.7 \times 0.91 \times 0.64}{0.49 \times 0.64 + 0.5} = -0.201$$

Instrument with a direct effect, $X = \pi Z + \rho W + v$, $Y = \beta X + \gamma Z + \lambda W + e$, with $\rho = \lambda = 0.5$, $\gamma = 0.05$:

$$\operatorname{plim}\, b_{IV} = \beta + \frac{\gamma}{\pi}, \qquad \operatorname{plim}\, b_{OLS} = \beta + \frac{\gamma\pi + \lambda\rho}{\pi^2 + \rho^2 + \sigma_v^2}$$

Symbolic checks: `scripts/glr_derivations.py`, six blocks, all PASS. Simulation (seed 2016): the omitted-variable and mediation identities hold exactly; the collider draw is within 0.03 of −0.201 (the asserted tolerance); 2SLS equals the reduced form over the first stage exactly. Back to the confounder, the collider, or the weak instrument.

GLR’s simulation: spurious instruments pass the usual tests

- **With β = 0 and three instruments that are noise or the regressor plus noise, IV returns 0.201, significant every time, rejects weak instruments every time, and passes the overidentification test 95.7 percent of the time.**

0.201

mean IV coefficient when the true β is 0 (close to ρ(X, ε) = 0.2)

100%

of replications: coefficient significant at 5 percent, and weak-instrument null rejected (statistic about 30)

95.7%

of replications: Sargan test fails to reject valid instruments

Design (p. 489): $y = X\beta + \varepsilon$ with $\beta = 0$, X and ε bivariate normal with unit variances and $\rho(X, \varepsilon) = 0.2$; instruments $z_1 = X + \eta_1$, $z_2 = \eta_2$, $z_3 = \eta_3$ with $\eta \sim N(0, 0.03^2)$; 1,000 replications of 1,000 firm-level observations. The instruments are strong by construction (z_1 is nearly X itself); the tests ask whether they are strong and whether they agree, not whether they are excludable.

> *“it is quite possible for completely spurious instruments to deliver bad inferences, yet easily pass tests for weak instruments and tests of overidentifying restrictions.”*
>
> Gow, Larcker, and Reiss (2016, p. 490)

GLR do not name the weak-instrument statistic or report the OLS mean separately. Back

A strategic auditor reverses the sign

- **On the same simulated data, the logit reports that equity pay reduces restatements while the structural estimate recovers the positive benefit coefficient that generated the data; the two are comparable in sign only.**

Estimate	Model	Coefficient on equity pay	Sign the theory implies
Logit, GLR Table 2 (p. 508)	nonstrategic auditor	**−0.504** (SE 0.130)	positive if $b_1 > 0$; the data have $b_1 = 10$
GMM, GLR Table 3 (p. 511)	strategic auditor, $\theta_6 = b_1/b_0$	**0.570** (SE 0.184), true value 0.500	positive

With a strategic auditor the restatement probability is decreasing in the benefit from misstatement (equation 12, p. 510): higher incentives make auditors work harder, so fewer misstatements survive to be restated. The logit describes the observed relation correctly; the structural model recovers the parameter the question asks about. Removing equity pay would raise the restatement rate by 24 percent, from 9.99 to 12.36 percent, because auditors reduce their effort (p. 514).

> *"if we used the wrong (nonstrategic) auditor structural model, we would have been led astray in our inferences about the effect of EQUITY."*
>
> Gow, Larcker, and Reiss (2016, p. 516)

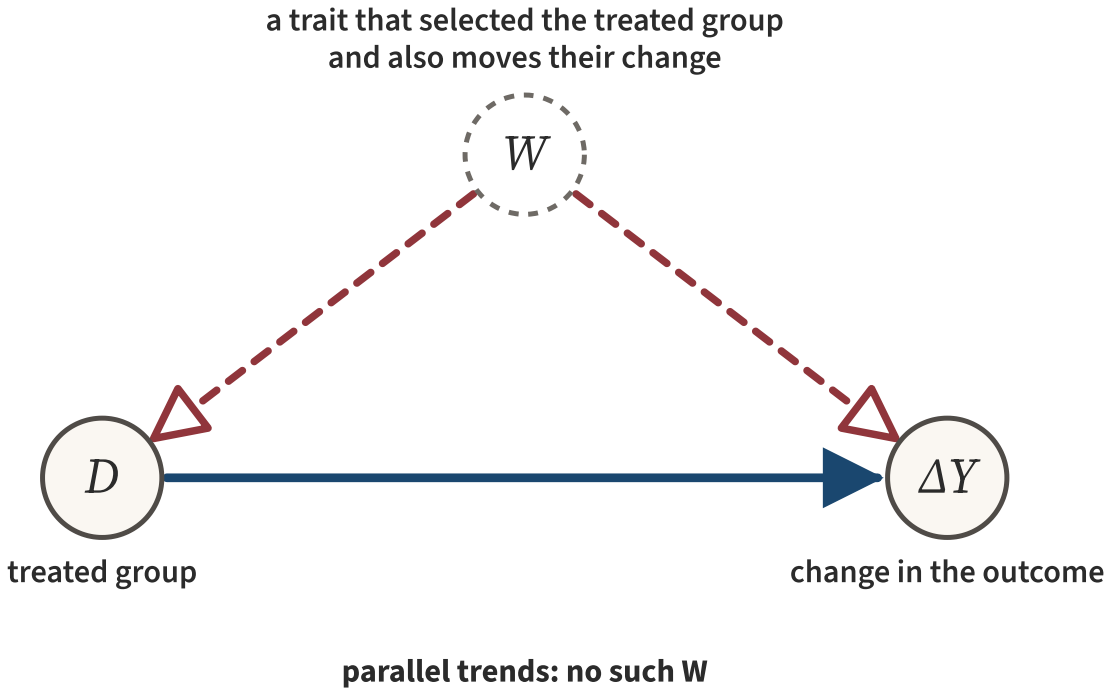
Back

What GLR leave out of difference-in-differences

- **GLR's DiD discussion is about assignment; treatment timing, staggered adoption, spillovers across units, and pre-trend plots are not discussed, and the seminar covers them elsewhere.**

- Not in the paper: staggered adoption or treatment timing, two-way fixed effects as a DiD estimator, spillovers or interference across units, pre-trend or placebo tests, synthetic control. "Parallel trends" appears only on p. 493.
- Week 5 (Schafhautle and Veenman 2024): parallel trends is a counterfactual assumption and is not testable; a pre-trend test can pass when the assumption fails and reject when it holds.
- Week 8 background reading: Baker, Larcker, and Wang (2022, *JFE*), on what staggered timing does to the two-way fixed effects estimate.

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A diagram for this note, not drawn by GLR. Dashed, hollow-headed: the arrows parallel trends assumes absent. With them present, the two-period DiD slope is $\tau + \theta\ \mathrm{Cov}(D, W)/\mathrm{Var}(D)$, where θ is the effect of W on the change in the outcome.